PRACTICAL WORK NO. 9. CONTROL OF TANK FILLING, MIXING AND PUMPING WITH FBD

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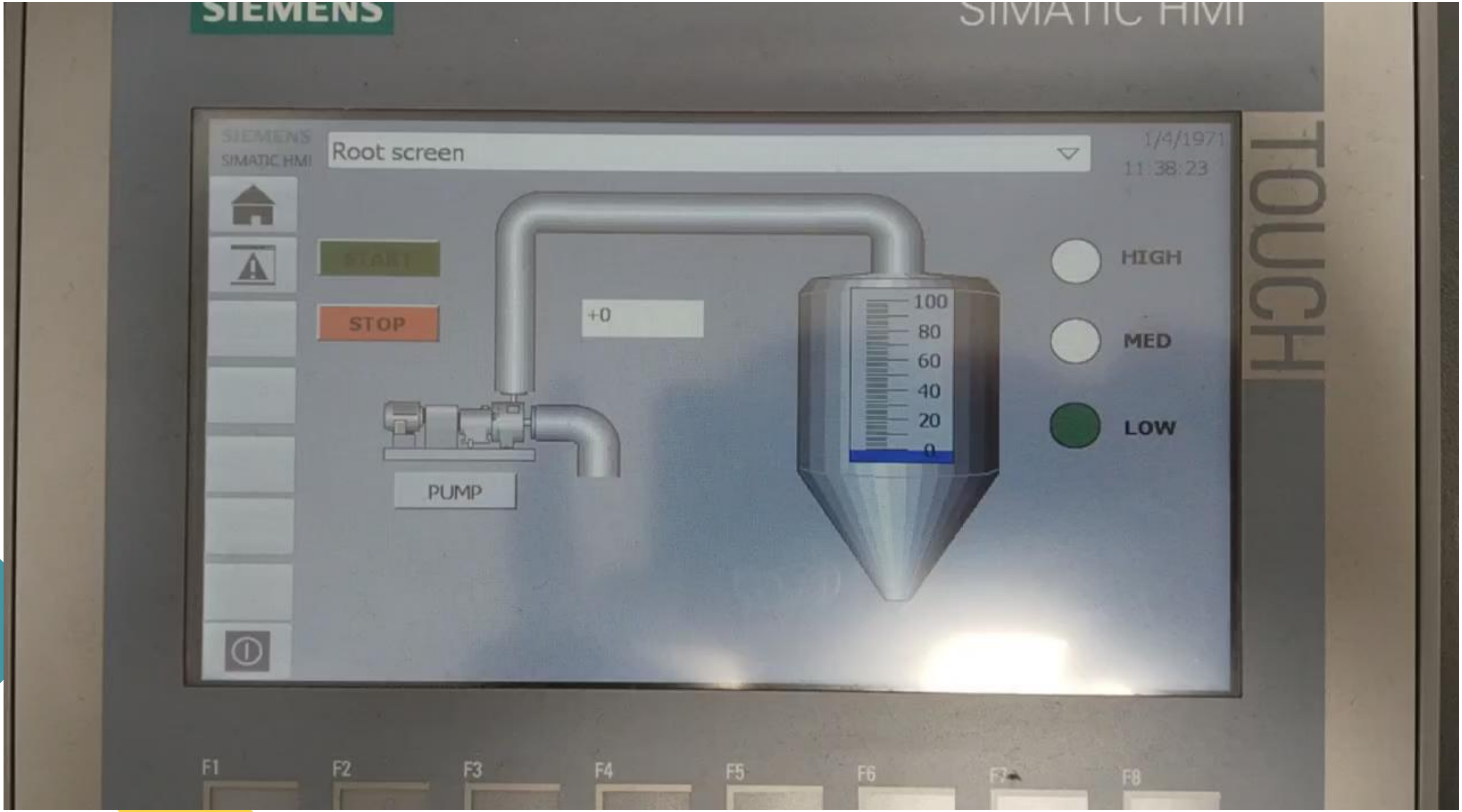
Task 1

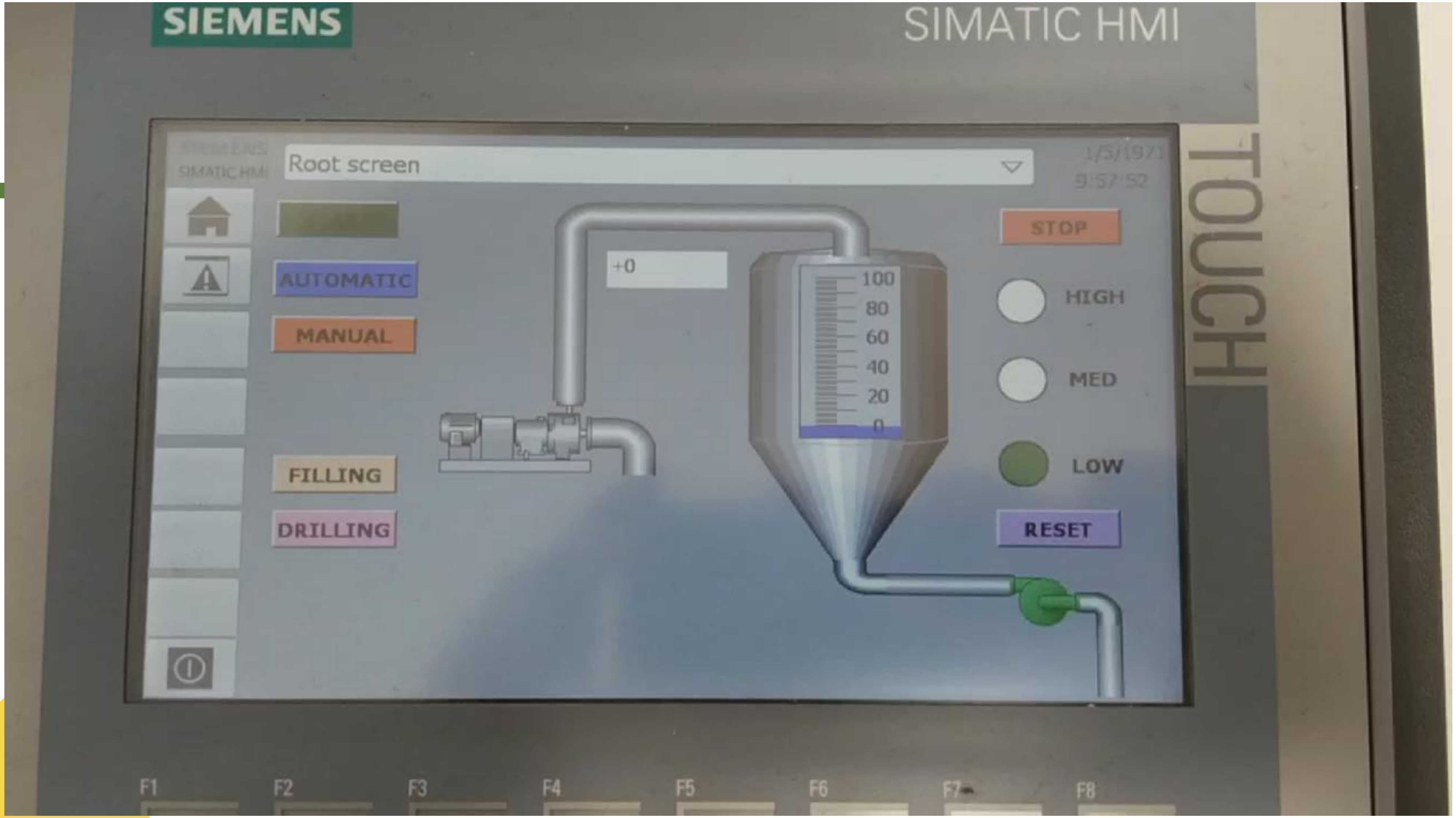
- Develop a program for automatic filling of the tank. When the set level thresholds are reached, the indicators LOW, MED, HIGH should be triggered, and the degree of filling should be displayed on a graphical scale. Implement visualization on the HMI: pump start/stop buttons, level indicators and filling scale.


SIEMENS
SIMATIC HMI
SIEMENS
SIMATIC HMI
Root screen
1/4/1971
11:38:23
TOUCH
STOP
+0
PUMP
100
80
60
40
20
0
HIGH
MED
LOW
F1
F2
F3
F4
F5
F6
F8

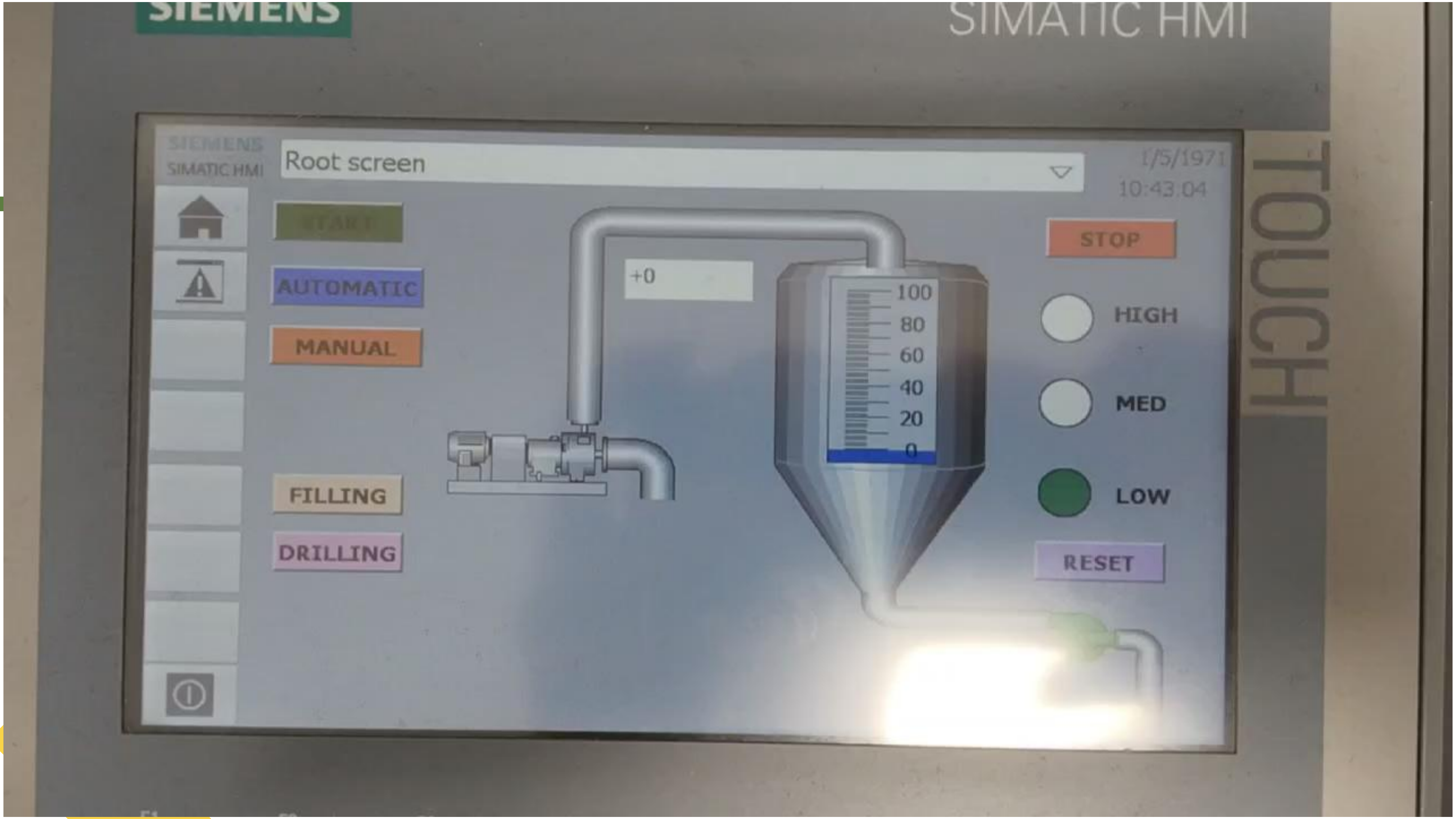
Task 2

- Develop a tank level control program with two modes: automatic and manual, as well as with two operation directions: filling and draining. The values "lower threshold" and "upper threshold" are set from the operator panel. In automatic mode "Filling", when the level is lower or equal to the lower threshold, switch on the pump and conduct the process until the upper threshold is reached, after which the pump is switched off. In automatic mode "Drain", when the level is above or equal to the upper threshold, open the drain valve and conduct the process until the level drops below the lower threshold, then switch off the pump. Level below the lower threshold, then close the valve. In manual mode the operator can switch on only one actuator - pump or drain valve; simultaneous switching on is prohibited. On the operator panel to provide buttons "Start" and "Stop", mode switches, input fields of thresholds, graphical scale of level and three indicators "low", "medium", "high". Provide emergency stop and correct operation of indication when the level passes the set thresholds.

Automatic


SIEMENS
SIMATIC HMI
Root screen
STOP
AUTOMATIC
MANUAL
+0
100
80
60
40
20
0
HIGH
MED
LOW
FILLING
DRILLING
RESET
TOUCH
F1
F2
F3
F4
F5
F6
F7
F8

Manual


SIEMENS
SIMATIC HMI
SIEMENS
SIMATIC HMI
Root screen
1/5/1971
10:43:04
AUTOMATIC
MANUAL
+0
100
80
60
40
20
0
STOP
HIGH
MED
LOW
RESET
FILLING
DRILLING
TOUCH

Conclusion

In this project, a complete automatic water tank control system was designed using a PLC and an HMI interface. The system can automatically fill the tank using a pump when the water level is low and empty it (simulated as “drilling”) when the level is high. All logic is implemented in the PLC using timers, sensors, and memory bits to safely control the process without overflow or dry running.

The HMI screen provides a user-friendly way to monitor and control the system. Operators can start/stop the process, switch between automatic and manual modes, and see the current water level as both a percentage and visual bar. Status indicators (LOW, MED, HIGH) and process labels (FILLING/DRILLING) make it easy to understand what the system is doing at any moment.